Fiscal Year:	FY 2015	**Task Last Updated:**	FY 10/15/2014
PI Name:	Matson, Douglas Ph.D.		
Project Title:	Electromagnetic Levitation Flight Support for Transient Observation of Nucleation Events		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Web Page:	https://		
City:	Medford	**State:**	MA
Zip Code:	02155	**Congressional District:**	7
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2009 ESA AO Biology on Sounding Rockets and Physical Sciences on All Platforms
Start Date:	12/01/2013	**End Date:**	11/30/2015
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	3	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Craven, Paul	**Contact Phone:**	256-544-7649
Contact Email:	pcravenRRR@nasa.gov		
Flight Program:	ISS		
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX14AB74G		
Performance Goal No.:			
Performance Goal Text:			
	Ed. NOTE (12/24/2013): Continuation of Electromagnetic Levitation Flight Support for Transient Observation of Nucleation Events (ELFSTONE) ; grant NNX10AV27G. See that project for previous reports. Funding is for Dr. Matson's role as Team Member on the subprojects described below, associated with ESA-sponsored main projects as noted. Main Project: Peritectic Alloy Rapid Solidification with Electromagnetic Convection (PARSEC); Science Coordinator is Dr. Thomas Volkmann, German Aerospace Center DLR, Institut für Materialphysik im Weltraum, Linder Höhe, Köln, Germany		

Task Description:	Subproject: Transformation kinetics - ISS Using results from thermophysical property evaluation and combining these with the MHD convection definition, we will set convection over as wide a range as is possible and conduct recalescence studies from the undercooled melt using the MSL-EML to compare to results from ground-based EML and ESL. The critical parameters of interest are evaluation of the incubation time between formation of metastable phases and subsequent transformation to the stable phase for Fe-based and titanium aluminide alloys. Since this transformation is very rapid, the high-speed video system must be used to image the processes associated with nucleation and growth during phase selection. Main project: Thermophysical Properties of Liquid Metallic Alloys - Modelling of Industrial Solidification Processes and Development of Advanced Products (THERMOLAB - ISS); Science Coordinator is Professor Hans-Joerg Fecht, Ulm University, Ulm, Germany Subprojects: Management and Coordination. Coordination of the scientific program, Coordination of the ground based measurement program, Coordination of the microgravity experiment program, Contact with the national and international space agencies. Short Duration Microgravity Measurements - PF. Conduct parabolic flights on selected alloys. Long Duration microgravity Experiments - ISS. Surface tension and viscosity as a function of temperature, Density as a function of temperature, Solidification growth velocity. Preparation of the Microgravity Experiments. Experimental protocols for processing on the ISS , Verification of flight samples for the ISS. MODELLING A: Thermodynamics, Phase Selection. Thermodynamic phase diagram calculation, Modeling of the influence of convection on phase selection.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The value of this research is that the results have application to the design of industrial welding, spray forming, and casting operations for a commercially important class of soft magnetic materials. In addition, this research addresses fundamental issues relating to rapid solidification behavior, metastable phase selection, and analysis of the processes governing microstructural evolution. Experiments focus on the use of convection as a processing control parameter.
Task Progress:	Conducted ESA parabolic flight 10-14 February 2014 in Merignac FR on FeCrNi flight alloy sample to test camera sensitivity and optimize operational settings for recalescence detection. Served as Symposium Organizer for American Society for Gravitational and Space Research (ASGSR) annual meeting to be held in Pasadena CA to be held on 21-26 October 2014. Invited speaker on nucleation kinetics for Dirk Holland-Moritz at the Herlach Symposium 29-30 September 2014 held at DLR in Köln Germany. International Working Group 1-2 October 2014 at DLR Köln Germany. IWG 10 involved recommending prioritization of the Oxygen Sensing and Control System over Batch 3 Ground Support, review of DLR staffing issues for flight support, and discussion of the impacts on the program management with the realignment of key ESA personnel. Conducted planning for flight support ESL quenching tests at ESL-MSFC on 6-17 October 2014. Plenary Speaker for 10th Asian Microgravity Society Symposium, 28 October 2014, Seoul ROK.
Bibliography Type:	Description: (Last Updated: 05/05/2026)
Articles in Peer-reviewed Journals	Lee J, Matson DM. "Prediction of mass evaporation of Fe50Co50 during measurements of thermophysical properties using an electrostatic levitator." Int J Thermophys. 2014 Oct;35:1697-704. https://doi.org/10.1007/s10765-014-1662-9 , Oct-2014